Cryogenic Requirements for a Compact Superconducting Quantum Computer

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Abstract

This report investigates the cryogenic requirements for operating a compact superconducting quantum computer based on transmon qubits. The Josephson-junction circuit is shown to provide the anharmonic energy spectrum required to define a two-level qubit, while thermal photon occupation motivates operation at temperatures of order $50mK$. The cryostat is analysed as a sequence of cooling stages: a pulse tube cryocooler, a continuous ${}^3He/{}^4He$ sorption cooler, and a miniature dilution refrigerator. Experimental temperature data demonstrates successful cooling of the mixing chamber to 66–$74mK$, close to the desired operating regime for superconducting qubits.

Contents

1 Introduction **2**

2 Superconducting Quantum Computing Platform **2**
2.1 Basic Quantum Information Theory 2
2.2 Josephson Junction 3
2.3 JC Circuit 6
2.4 Transmon Qubits 7
2.5 Operational Requirements 9

3 Our Cryostat **10**
3.1 Pulse Tube Cryocooler ($293K \rightarrow 4K$) 10
3.2 $^{3/4}He$ Sorption Cooler ($4K \rightarrow 0.3K$) 11
3.2.1 Volume Estimation 13
3.3 Split Condenser 15
3.4 Miniature Dilution Refrigerator ($0.3K \rightarrow 0.05K$) 16

4 Conclusion **20**

References **21**

1 Introduction

Quantum computing has emerged as one of the most important technological challenges of modern physics and engineering, motivated by the possibility that quantum systems may outperform classical computers for specific computational tasks. A major catalyst for quantum computing development came from the work of Peter Shor, whose Shor's algorithm demonstrated in 1994 that a sufficiently powerful quantum computer could factorise large integers exponentially faster than known classical methods [1]. This has important implications for modern cryptographic systems such as RSA cryptosystem, whose security relies on the practical difficulty of prime factorisation [2]. In response to these challenges, quantum communication protocols such as the BB84 protocol have been developed, using fundamentally quantum mechanical effects to enable secure key distribution and interception detection [3]. Experimental demonstrations of large-scale quantum-secure communication networks have already been achieved, including the Micius satellite quantum key distribution experiments in 2017 [4] and the Beijing–Shanghai Quantum Backbone [5], demonstrating that quantum-secure communication is already technologically feasible and may provide protection against future quantum-enabled cryptographic attacks.

Several physical platforms exist for implementing qubits, including trapped ions, photonic systems, spin qubits, and superconducting circuits. Among these, superconducting qubits have become one of the leading architectures due to their scalability, compatibility with modern microfabrication techniques, and rapid experimental progress. In particular, the transmon qubit, developed from the Josephson junction circuit, has become the dominant superconducting qubit design used by companies such as IBM and Google. Modern superconducting quantum processors now contain hundreds to over a thousand qubits, with continuing advances in coherence times and error correction. For example, IBM Condor, released in 2023, contains $1,121$ superconducting transmon qubits [6]. Despite these advances, current devices remain within the noisy intermediate-scale quantum (NISQ) era, where decoherence, gate errors, and limited error correction continue to restrict large-scale fault-tolerant quantum computation [7].

A fundamental requirement of superconducting quantum computation is cryogenic operation. Transmon qubits typically operate at transition frequencies of order $10GHz$ [8], corresponding to thermal energy scales of order $100mK$. And to suppress thermal excitations and environmental noise, qubits must operate at temperatures of $\sim 50mK$, as is shown in Section 2.5. Achieving such temperatures requires sophisticated cryogenic systems combining multiple refrigeration stages, including pulse tube cryocoolers, sorption coolers, and dilution refrigerators. Developments in low-temperature physics and cryogenic engineering have therefore become essential to the advancement of scalable quantum computing technologies. Modern cryogenic research aims to develop systems with lower base temperatures, higher cooling powers, reduced vibrations, and more compact designs. Progress in this field is driven not only by quantum computing, but also by applications in healthcare, particle physics, nuclear fusion, space science, and aerospace engineering.

This report investigates the physical requirements for operating a compact superconducting quantum computer based on transmon qubits. The theory of superconducting qubits and Josephson junction circuits is first introduced, before analysing the cryogenic systems required to achieve millikelvin operating temperatures. In particular, the operation of pulse tube cryocoolers, helium sorption coolers, and miniature dilution refrigerators is examined, alongside the thermodynamic and quantum mechanical principles underlying their performance.

2 Superconducting Quantum Computing Platform

2.1 Basic Quantum Information Theory

As discussed in the Introduction 1 quantum computers have the potential to provide huge advances in computational power, tackling problems previously unattainable to a classical computer. But how do they work? Well in comparison to a classical computer whose fundamental building blocks are billions of switches, transistors. Which are mathematically represented as a bit, which can take only the discrete state 0 or 1 not both at the same time. Quantum bits (qubits) are the mathematical

representation for the building blocks of quantum computers. But they are fundamentally different mathematical objects, qubits are described by state vectors in a two-dimensional complex Hilbert space $\mathcal{H} \cong \mathbb{C}^2$. And can hence be in a superposition of both states at the same time. Choosing the computational basis $\{|0\rangle, |1\rangle\}$, an arbitrary pure state, $|\psi\rangle \in \mathcal{H}$ can be written as

$$|\psi(t)\rangle = \alpha(t)|0\rangle + \beta(t)|1\rangle, \quad \text{where } \langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

The second equation is the normalisation constraint that $\alpha, \beta \in \mathbb{C}$ satisfy. These coefficients are probability amplitudes, with measurement in the computational basis yielding outcomes $|0\rangle$ and $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

This is all nice, but how do we create our physical quantum system that can represent the qubits, ie. the equivalent of a transistor. Well in practice, the computational basis $\{|0\rangle, |1\rangle\}$ is just a two-dimensional subspace of the Hilbert space $\{|n\rangle\}$ for the quantum system with dynamics governed by unitary time evolution. In contrast to classical bits, which do not possess intrinsic continuous-time dynamical laws, the state $|\psi\rangle \in \{|n\rangle\}$ of a closed quantum system evolves according to the Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle \implies |\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-i\hat{H}t/\hbar}.$$

where $\hat{H}$ is the Hamiltonian of whatever quantum system you choose, representing the total energy observable. Its eigenvalues E_n correspond to the allowed energy levels of the system, defined by

$$\hat{H}|n\rangle = E_n|n\rangle.$$

The differences between energy eigenvalues $(E_{n+1} - E_n)$ determines the characteristic transition frequencies $\omega_{(n+1)(n)}$ of the system. These are exploited to selectively manipulate our computational basis, by for example exciting from $|0\rangle \to |1\rangle$ driven by a photon with transition frequency ω_{10}. This requires the spectrum of $\hat{H}$ to be anharmonic, so that

$$\hbar\omega_{01} = E_1 - E_0 \neq E_2 - E_1 = \hbar\omega_{12}, \tag{1}$$

ensuring that transitions can be selectively driven without populating higher levels. Otherwise if $\omega_{10} = \omega_{21}$, using the same afore mentioned example, one could have a transition from $|1\rangle \to |2\rangle$ using the same photon that was intended to excite $|0\rangle \to |1\rangle$. And hence creating a system that doesn't stay in our intended computational basis, $\{|0\rangle, |1\rangle\}$.

An example of a quantum system that can exhibit this required anharmonicity is a circuit containing a Josephson junction which can create a computational basis known as superconducting qubits. The non-linear inductive energy of Josephson junctions leads to a Hamiltonian of the form [8],

$$\hat{H} = 4E_C\hat{n}^2 - E_J\cos\hat{\phi}, \quad [\hat{\phi}, \hat{n}] = i, \tag{2}$$

where E_C and E_J are the charging and Josephson energies, respectively. Quantisation of this circuit yields a discrete, anharmonic spectrum, from which a two-level subspace can be selected.

2.2 Josephson Junction

As mentioned in the previous Section 2.1, we will be looking at circuits containing a Josephson Junction. A Josephson Junction contains an insulating barrier separating two superconducting wires. Superconductivity is a quantum phase of matter that occurs at very low temperatures (for Aluminium $T_c \approx 1.2K$ [9]) where electrons have zero resistivity. These freely-moving electrons move through the lattice and polarise the ionic background, this lattice distortion (a phonon) can attract a second electron, leading to an effective electron–electron attraction and the formation of a Cooper pair [10]. Consequently, Cooper pairs have an effective charge q and mass m^* of $-2e, 2m_e$ respectively and an integer spin which means these cooper pairs behave like bosons. And at superconducting temperatures all the cooper pairs condense into the same quantum state described by a macroscopic wavefunction,

$$\psi(\mathbf{r}, t) = \sqrt{n_s}\exp(i\theta(\mathbf{r}, t)),$$

where n_s is the number density of cooper pairs, $\theta(\mathbf{r}, t)$ is the macroscopic phase. Just like with the classic example of a particle in a finite potential well, the quantum mechanical result shows that there is a non-zero probability of finding the particle outside the barrier of the well. This is the same with a Josephson junction where given two superconducting wires separated by an insulating barrier, if they are close enough, there is a non-zero probability of finding cooper pairs in the insulating barrier. The cooper pairs can hence tunnel across the insulating barrier, creating a current. This effect is best demonstrated by Diagram 1a.

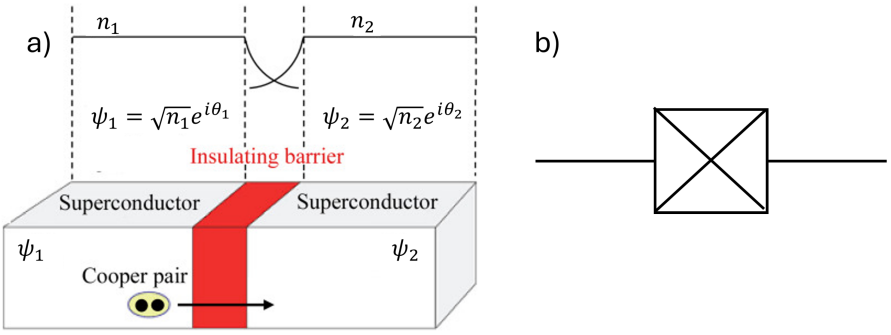


Figure 1: a) Cooper pairs tunnelling through an insulating barrier in between 2 superconducting wires, a Josephson junction, adapted from [11]. b) Josephson junction circuit symbol.

This current can be calculated using the following setup. Each wire is described by a macroscopic wavefunction and both evolve according to the coupled Schrödinger equation (with coupling strength K) [12],

$$\psi_1(\mathbf{r}, t) = \sqrt{n_1}\exp(i\theta_1(\mathbf{r}, t)), \qquad i\hbar\frac{d\psi_1}{dt} = E_1\psi_1 + K\psi_2, \tag{3}$$

$$\psi_2(\mathbf{r}, t) = \sqrt{n_2}\exp(i\theta_2(\mathbf{r}, t)), \qquad i\hbar\frac{d\psi_2}{dt} = E_2\psi_2 + K\psi_1. \tag{4}$$

The first step is to get I_N, the rate at which Cooper pairs cross the junction, from the integrated continuity equation

$$\frac{\partial}{\partial t}\int_V \rho dV = -\oint_{\partial V}\mathbf{j}\cdot d\mathbf{S} \implies \frac{\partial}{\partial t}\int_V |\psi_1(\mathbf{r}, t)|^2 dV = -I_N.$$

where ρ is the probability density and the relationship between the probability density and the wavefunction ($\rho = |\psi_1(\mathbf{r}, t)|^2$) is the Born rule. If we make the single-mode approximation ($\psi_1(\mathbf{r}, t) \approx \psi_1(t)$). This approximation can be justified within Ginzburg-Landau theory, a phenomenological (macroscopic) theory of superconductivity. Where the defining equation of the theory is the free energy functional neglecting vector potential term [13],

$$f_{\mathrm{GL}}[\psi] = \int d^3r(\alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{\hbar^2}{2m^*}|\nabla\psi|^2).$$

When minimizing this functional, the gradient term $|\nabla\psi|^2 \sim \frac{|\psi(\mathbf{r}+\epsilon)-\psi(\mathbf{r})|^2}{\epsilon}$ is large if $\psi(\mathbf{r})$ varies rapidly. Hence this term energetically penalizes spatial variations of the wavefunction. As a result, the free energy is minimized when ψ is approximately uniform over length scales smaller than the coherence length ξ, which characterizes the scale over which ψ can vary appreciably [14]. This approximation leads to,

$$\implies I_N = -\frac{\partial}{\partial t}\int_V |\psi_1(\mathbf{r}, t)|^2 dV \approx -V_1\frac{\partial|\psi_1(t)|^2}{\partial t}. \tag{5}$$

Substituting the Schrödinger equations (3) and (4) into the time derivative term,

$$\frac{\partial|\psi_1|^2}{\partial t} = \frac{\partial(\psi_1^*\psi_1)}{\partial t} = \dot{\psi}_1^*\psi_1 + \psi_1^*\dot{\psi}_1,$$

$$= \frac{i}{\hbar}(E_1\psi_1^* + K\psi_2^*)\psi_1 + \psi_1^*\left(-\frac{i}{\hbar}(E_1\psi_1 + K\psi_2)\right),$$

$$= \frac{iK}{\hbar}(\psi_2^*\psi_1 - \psi_1^*\psi_2),$$

$$= -\frac{2K}{\hbar}\,\mathrm{Im}(\psi_2^*\psi_1). \tag{6}$$

Using the wave functions (3) and (4) we can calculate an expression for the imaginary part,

$$\psi_1^*\psi_2 = \sqrt{n_1 n_2}\, e^{i(\theta_2-\theta_1)} \implies \mathrm{Im}(\psi_1^*\psi_2) = \sqrt{n_1 n_2}\sin(\theta_2-\theta_1). \tag{7}$$

Using equations (5), (6) and (7) we can calculate the electrical current. Which is the rate at which Cooper pairs cross the junction times the charge of each copper pair $q = -2e$,

$$I = -(-2e)I_N = -2eV_1\frac{\partial|\psi_1|^2}{\partial t} = \frac{4eKV_1}{\hbar}\,\mathrm{Im}(\psi_2^*\psi_1),$$

$$= I_c\sin(\theta_2-\theta_1) = I_c\sin(\Delta\theta), \quad I_c = \frac{4eKV_1\sqrt{n_1n_2}}{\hbar}, \tag{8}$$

where the first minus sign comes from the convention that current flows in the opposite direction to electron (Cooper pair) flow. $I_c\sin(\Delta\theta)$ is known as the Josephson Effect. To calculate $\Delta\theta$, we substitute the macroscopic wavefunctions into their coupled Schrödinger equations (3) and (4),

$$i\hbar\frac{d\psi_1}{dt} = i\hbar(i\frac{d\theta_1}{dt})\psi_1 = E_1\psi_1 + K\psi_2,$$

$$i\hbar\frac{d\psi_2}{dt} = i\hbar(i\frac{d\theta_2}{dt})\psi_2 = E_2\psi_2 + K\psi_1.$$

If we solve these simultaneous equations, remembering that ψ_1 and ψ_2 are independent we get,

$$\frac{d\theta_1}{dt} = \frac{K-E_1}{\hbar}, \quad \frac{d\theta_2}{dt} = \frac{K-E_2}{\hbar},$$

$$\implies \frac{d\theta_1}{dt} - \frac{d\theta_2}{dt} = \frac{d\Delta\theta}{dt} = \frac{E_2-E_1}{\hbar} = \frac{2eV}{\hbar}.$$

$$\therefore \Delta\theta = \Delta\theta_0 + \frac{2eV}{\hbar}t, \tag{9}$$

where V is the voltage across the insulating barrier and is given by $E_2 - E_1 = 2eV$. This shows that a constant non-zero voltage across a Josephson junction will induce an AC superconducting current [12].

To calculate the energy stored in a Josephson junction, which gives the non-linear term $\cos\hat{\phi}$ in the hamiltonian mentioned in the Quantum Information Theory Section 2.1, use the relationship between phase and magnetic flux $\Phi = \frac{\hbar}{2e}\Delta\theta$ [12], we obtain $V = \frac{d\Phi}{dt}$ using equation (9). Which is Faraday's law. This allows the Josephson junction to be interpreted as an effective inductive element. The energy stored in a Josephson junction can be calculated using the power law for electric circuits $P = IV$ and the Josephson effect (8),

$$E = \int I(t)V(t)dt = \int I_c\sin\left(2\pi\frac{\Phi}{\Phi_0}\right)\frac{d\Phi}{dt}dt = -E_J\cos\left(2\pi\frac{\Phi}{\Phi_0}\right), \tag{10}$$

where the Josephson energy is $E_J = \frac{\Phi_0 I_c}{2\pi}$ and the magnetic flux quanta is $\Phi_0 = \frac{\pi\hbar}{e}$.

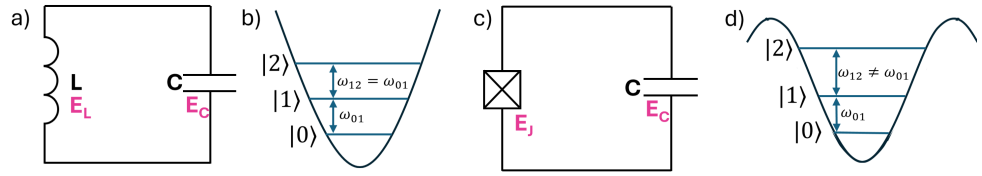


Figure 2: a) LC circuit, b) Harmonic Potential, c) JC circuit, d) Anharmonic Potential. Adapted from [15].

2.3 JC Circuit

The aim of this Section is to recover the non-linear hamiltonian (2) from the Quantum Information Theory section 2.1 and to show its anharmonicity. To do this we will look at a simple JC circuit, this consists of the Josephson junction shunted by a capacitor, see Image 2c.

To understand how we can get the non-linear hamiltonian first consider the simple LC circuit, Image 2a, the total energy for this circuit is,

$$H = E_C + E_L = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}, \tag{11}$$

where E_C, Q, C is the energy, total charge and capacitance of the capacitor, and E_L, Φ, L is the energy, total magnetic flux and inductance of the inductor. If we promote the classical circuit variables Q, Φ to quantum operators $\hat{Q}, \hat{\Phi}$, canonical quantisation maps the classical phase space (Q, Φ), equipped with the Poisson bracket $\{\Phi, Q\} = 1$, to operators on a Hilbert space satisfying the canonical commutation relation $[\hat{\Phi}, \hat{Q}] = i\hbar$. We get a hamiltonian with the same form as the quantum harmonic oscillator (QHO),

$$\hat{H}_{LC} = \frac{\hat{Q}^2}{2C} + \frac{1}{2}\frac{1}{L}\hat{\Phi}^2 \quad \leftrightarrow \quad \hat{H}_{QHO} = \frac{\hat{p}^2}{2m} + \underbrace{\frac{1}{2}m\omega^2\hat{x}^2}_{\text{quadratic potential}} . \tag{12}$$

And as $\hat{p}, \hat{x}$ also have the same commutation relations as $\hat{Q}, \hat{\Phi}$ with $[\hat{x}, \hat{p}] = i\hbar$. We can use the standard result from the QHO for the energy levels E_n,

$$\hat{H}|n\rangle = E_n|n\rangle \implies E_n = \hbar\omega(n + \frac{1}{2}),$$

where $\omega = \frac{1}{\sqrt{LC}}$, due to $m = C$ and $m\omega^2 = \frac{1}{L}$. Note that all transitions have the same frequency, see Image 2b, excitations from $|0\rangle \to |1\rangle$ also excites $|1\rangle \to |2\rangle$. As mentioned in the Quantum Information section 2.1 this is not ideal for a qubit. To get the hamiltonian for the JC circuit we replace the inductive term in the LC hamiltonian (12) with the energy in the Josephson junction (10) [16],

$$\hat{H} = \frac{\hat{Q}^2}{2C} \underbrace{- E_J \cos\left(2\pi\frac{\hat{\Phi}}{\Phi_0}\right)}_{\text{cosine potential}} .$$

Redefining the charge operator in terms of cooper pair number operator ($\hat{Q} = 2e\hat{n}$) and the flux operator as the phase operator ($2\pi\frac{\hat{\Phi}}{\Phi_0} = \frac{\hat{\phi}}{\hbar}$), these definitions also retain the commutation relations $[\hat{\phi}, \hat{n}] = i\hbar$. The hamiltonian becomes,

$$\hat{H} = 4E_C\hat{n}^2 - E_J \cos\frac{\hat{\phi}}{\hbar},$$

where the charging energy $E_C = \frac{e^2}{2C}$. Taylor expanding the cosine and ignoring the constant energy offset $-E_J$, as this will not affect the energy spacing and hence transition frequencies,

$$\hat{H} = \underbrace{4E_C\hat{n}^2 + \frac{1}{2}\frac{E_J}{\hbar^2}\hat{\phi}^2}_{\hat{H}^{(0)}=QHO} - \underbrace{\frac{E_J}{24}(\frac{\hat{\phi}}{\hbar})^4}_{\hat{H}^{(1)}=Pertubation} + O(\hat{\phi}^6).$$

Later in section 2.4 we show $E_J >> E_C$, the transmon regime (18). This means the cosine potential dominates and forms a deep, steep potential well. The low-energy states are therefore strongly confined near the minimum at $\phi = 0$, meaning the phase fluctuations are small ($\phi << 1$). This means we can ignore high order terms. Treating the quartic term as a perturbation to the QHO, first-order perturbation theory gives,

$$(\hat{H}^{(0)} - \hat{H}^{(1)})|n\rangle \approx (E_n - \Delta E_n)|n\rangle, \quad \text{with } E_n = \hbar\omega_p(n + \frac{1}{2}), \tag{13}$$

where $\omega_p = \frac{\sqrt{8E_C E_J}}{\hbar}$, due to $\frac{1}{2m} = 4E_C$ and $m\omega^2 = \frac{E_J}{\hbar^2}$. To calculate ΔE_n requires a little more effort. We can use the annihilation $\hat{a}$ and creation $\hat{a}^\dagger$ operators and their relation with the $\hat{x}$ operator,

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a}) \implies \hat{\phi} = \hbar(\frac{2E_C}{E_J})^{\frac{1}{4}}(\hat{a}^\dagger + \hat{a}),$$

$$\implies \Delta E_n = \langle n|\hat{H}^{(1)}|n\rangle = \langle n|\frac{E_J}{24}(\frac{\phi}{\hbar})^4|n\rangle = \frac{E_C}{12}\langle n|(\hat{a}^\dagger + \hat{a})^4|n\rangle,$$

$$(\hat{a}^\dagger + \hat{a})^4 = \hat{a}^4 + (\hat{a}^\dagger)^4 + 4(\hat{a}^\dagger)^3\hat{a} + 6(\hat{a}^\dagger)^2\hat{a}^2 + 12\hat{a}^\dagger\hat{a} + 4\hat{a}^\dagger\hat{a}^3 + 6((\hat{a}^\dagger)^2 + \hat{a}^2) + 3,$$

where we have used $[\hat{a}, \hat{a}^\dagger] = 1$ to simplify the expansion. Only the terms symmetric in powers of $\hat{a}^\dagger, \hat{a}$ remain when sandwiched between $|n\rangle$, these terms include $6(\hat{a}^\dagger)^2\hat{a}^2, 12\hat{a}^\dagger\hat{a}, 3$,

$$\implies \Delta E_n = \frac{E_C}{12}\langle n|\big(6(\hat{a}^\dagger)^2\hat{a}^2 + 12\hat{a}^\dagger\hat{a} + 3\big)|n\rangle,$$

$$= \frac{E_C}{2}(n^2 + n + \frac{1}{2}), \tag{14}$$

where we have used $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$. We can now calculate the transition frequencies from $|0\rangle \to |1\rangle$ and $|1\rangle \to |2\rangle$ to show the anharmonicity required from equation (1) [16],

$$\hbar\omega_{01} = (E_1 - \Delta E_1) - (E_0 - \Delta E_0) = \sqrt{8E_C E_J} - E_C, \tag{15}$$

$$\hbar\omega_{12} = (E_2 - \Delta E_2) - (E_1 - \Delta E_1) = \sqrt{8E_C E_J} - 2E_C,$$

$$\implies \hbar\omega_{01} \neq \hbar\omega_{12}. \tag{16}$$

2.4 Transmon Qubits

A transmon qubit is a superconducting quantum circuit built using a modified JC circuit and other components. Adding an extra Josephson junction in parallel to the existing one in the JC circuit makes a transmon, forming a SQUID (superconducting quantum interference device) loop [17]. This also allows us to tune the qubit frequency which is very useful for controlling interactions with other qubits and for coupling to resonators. A resonator is a component used to control and read out the state of the qubit [18]. This can be done because now each Josephson junction in our SQUID loop has a phase drop $\Delta\theta_1, \Delta\theta_2$ which is tuned by an externally applied magnetic flux Φ_{ext} [19],

$$\Delta\theta_1 - \Delta\theta_2 = 2\pi\frac{\Phi_{\text{ext}}}{\Phi_0}.$$

Using the Josephson effect (8), the total current becomes,

$$I_{\text{tot}} = I_c \sin\Delta\theta_1 + I_c \sin\Delta\theta_2 \implies I_{\text{tot}} = \underbrace{2I_c \cos\left(\frac{\pi\Phi_{\text{ext}}}{\Phi_0}\right)}_{I_c^{\text{eff}}} \sin(\Delta\theta_{\text{avg}}). \tag{17}$$

Using the same maths from section 2.2 to derive the energy stored in a Josephson junction (10), if we replace $I(t)$ with the new total current (17), we hence modify the previous Josephson energy E_J term such that,

$$E_J^{\text{eff}} = 2E_J \cos\left(\frac{\pi\Phi_{\text{ext}}}{\Phi_0}\right).$$

And as the Josephson energy is related to $\omega_{01}(E_J)$ via equation (15), we can now tune the qubit frequency using an external magnetic field $\omega_{01}(E_J^{\text{eff}})$.

The addition of a gate capacitor C_g coupled to the transmon qubit allows us to model the effect of environmental charge fluctuations, such as stray electrostatic charges and background charge noise. The gate capacitor is biased by a gate voltage V_g, which induces an offset charge. Fluctuations in this offset charge, due to randomly discharged cooper pairs from the gate capacitor, mimic the effect of random environmental charge noise acting on the qubit. This modifies the charging energy of the circuit and introduces an offset charge term [8],

$$n_g = \frac{C_g V_g}{2e}.$$

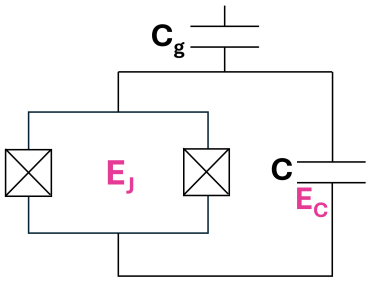


Figure 3: Circuit diagram of a Transmon coupled to a gate capacitor. Adapted from [20].

See Figure 3 for a circuit diagram of these additional components. This offset charge term modifies the Hamiltonian from (2),

$$\hat{H} = 4E_C\,(\hat{n} - \hat{n}_g)^2 - E_J \cos\hat{\phi} \quad \text{with } [\hat{\phi}, \hat{n}] = i.$$

Solving this hamiltonian can be done using similar techniques as in section 2.3. But we can also solve it computationally and plot a graph of the energy levels E_n against gate charge count n_g for different ratios of E_J/E_C, see plot 4 [21].

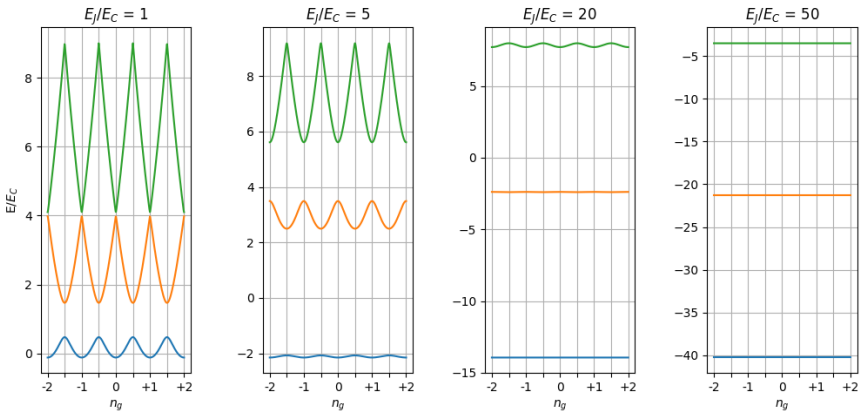


Figure 4: Plot of Energy levels E_n for $n = 0, 1, 2$ against gate charge count n_g for different ratios of E_J/E_C. Plot created using [21].

This plot shows, for small ratios of E_J/E_C, the energy eigenvalues E_n vary strongly with the offset gate charge n_g. Consequently, the qubit becomes highly sensitive to charge fluctuations in the environment, leading to significant charge noise. However, as the ratio E_J/E_C increases, the energy levels become increasingly flat with respect to n_g, greatly suppressing charge sensitivity.

At the same time, increasing E_J/E_C causes the energy spectrum to become progressively more harmonic, such that the transition frequencies between neighbouring states become increasingly similar. This corresponds to a reduction in anharmonicity, as can be observed for $E_J/E_C = 50$. As

has been discussed, this affects the selective control of the qubit which is undesirable. Therefore the transmon regime is defined by [22]

$$\frac{E_J}{E_C} \gg 1, \tag{18}$$

where charge sensitivity is exponentially suppressed while sufficient anharmonicity is retained for reliable qubit operation.

2.5 Operational Requirements

Equation (16) from Section 2.3 demonstrates the anharmonicity required for qubit operation, individual transitions can now be selectively addressed. For a typical transmon qubit, the $|0\rangle \to |1\rangle$ transition frequency is of order $f_{01} \sim 6GHz$ [8]. This corresponds to a thermal energy scale,

$$\hbar\omega_{01} \sim k_B T \implies \frac{h f_{01}}{k_B} \approx 300mK,$$

meaning thermal energies above this scale would significantly populate excited states. Since quantum computation generally requires the qubit to be initialized in its ground state $|0\rangle$, transmon qubits must therefore operate at temperatures well below this limit.

The characteristic timescale over which an excited qubit decays from $|1\rangle \to |0\rangle$ is known as the relaxation time T_1, and maximizing this time is a major goal in quantum computer design [23]. Although most cryogenics systems can cool down to $T < 300mK$, we actually need even lower temperatures of $\sim 50mK$. The reason for this is due to residual thermal photons still in the environment that can excite the qubit [24]. The average thermal photon occupation number for a single EM mode of frequency ω is given by the Bose-Einstein distribution,

$$\bar{n}(f,T) = \frac{1}{e^{hf/k_B T} - 1}.$$

There are many modes in the environment, e.g. on the resonator, which will have a different mode where thermal fluctuations will resonate. And since transmon qubits couple most strongly to modes near their transition frequency ω_{01}, thermal photons at this frequency can excite the qubit. See Figure 5 for a plot of this distribution at transition frequency $f = f_{01} = 6GHz$. As can be seen in this plot the photon occupation decreases rapidly only at temperatures below $50mK$, motivating the use of ultra-low-temperature cryogenic systems.

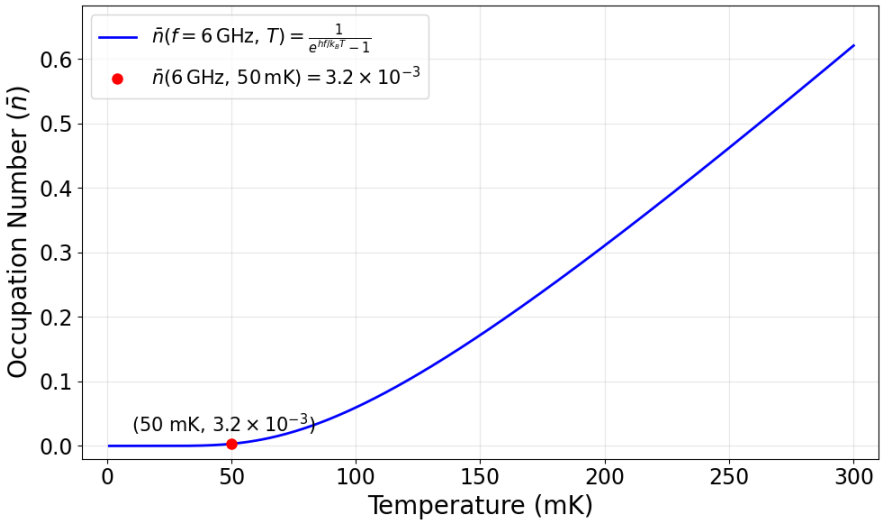


Figure 5: Bose-Einstein Occupation number vs Temperature for a thermal photon with frequency $f = f_{01} = 6GHz$.

3 Our Cryostat

As explained in Section 2.5, we need temperatures $\sim 50mK$ to stop thermal photons exciting our qubit. To accomplish this we use a cooling system which encloses the thermal load of our quantum computer. Our cooling system is called a cryostat which cools things down to cryogenic temperatures (typically temperatures less than $120K$ [25]). Our cryostat is made up of three main components, a pulse tube cryocooler (PTC) which cools from $273K \rightarrow 4K$, a 2-stage continuous sorption cooler $4K \rightarrow 0.3K$ and a miniature dilution refrigerator (mini-DR) $0.3K \rightarrow 0.05K$, obtaining the desired temperature of $50mK$. Both the 2-stage continuous sorption cooler and mini-DR require the quoted starting temperatures to actually operate, both do not work at room temperature. As a result we need to thermally link the coldest part of the PTC to the 2-stage sorption cooler and the coldest part of the 2-stage sorption cooler to the mini-DR. Thermally linked means the components are connected in a way that allows heat to flow efficiently, typically via conduction,

$$\dot{Q} \propto \kappa A \frac{\Delta T}{\Delta x},$$

so to maximise this heat flow from e.g. $T_A \rightarrow T_B$ we require large contact area A and the 2 heat sources connected via short distances Δx. But also more importantly in the case of our cryogenics, to choose a material with a high thermal conductivity κ. Hence between components we choose copper plates which has a very high thermal conductivity (at low $T, \kappa \sim 10^3 Wm^{-1}K^{-1}$ [26]) compared to most commercial materials. But equally in some cases we want to prevent heat flow, such as in the support rods and cryostat structural frame. If we had used copper supports between room temperature and the $50mK$ plate of the mini-DR, enormous heat would enter from the outside. So for these parts of the cryostat we use stainless steel which has a very low thermal conductivity (at low $T, \kappa \sim 1 Wm^{-1}K^{-1}$ [27]) and hence acts like a thermal insulator. So copper is used when you want parts to quickly reach the same temperature and stainless steel when you don't.

3.1 Pulse Tube Cryocooler ($293K \rightarrow 4K$)

The first component in our cryostat. The pulse tube cryocooler (PTC) cools the copper plate down to $4K$. A simple PTC has 7 main components. A compressor, heat exchangers, regenerator, pulse tube, needle valve, inertance tube and buffer tank. Where the needle valve, inertance tube and buffer tank can be collectively called the delay mechanism. The main mechanism that produces cooling is the periodic expansion and compression of gas [28].

A simple thermodynamic model describing how expansion and compression cycles cause cooling, albeit no net cooling. For a piston compressing a fixed volume box, the 1st law of thermodynamics says,

$$dU = dQ - PdV,$$

where P is the pressure exerted by the piston. In the compression stage there is no heat flow ($dQ = 0$) due to being a quick process and a well insulated system. Using the internal energy of an ideal gas ($dU = C_V dT$) we get,

$$C_V dT = -PdV.$$

So as the piston expands $dV \uparrow \Rightarrow dT \downarrow$, and similarly when it compresses $dV \downarrow \Rightarrow dT \uparrow$. The expansion stage is what causes cooling, but then the compression stage exactly cancels out that cooling. For a continuous PTC we can't just expand a system indefinitely to produce cooling, this is physically impossible. We have to compress to start the expansion stage again. A PTC fixes this problem using careful placement of additional components. First in the compression stage it vents out as much "hot" heat as it can to the environment ($dQ_c < 0$) and in the expansion stage it stores as much "cold" heat as possible ($dQ_e > 0$) so that $C_V dT_c = -PdV_c + dQ_c < -PdV_e + dQ_e = C_V dT_e \Rightarrow dT_c < dT_e$. This means, if at the beginning of the cycle we have temperature T_i. Then during compression we have temperature $T_c = T_i + dT_c$ and after expansion we have temperature $T_e = T_c - dT_e$. And hence $T_e < T_i$, overall a net cooling. The components that vent out and store heat are called hot (HHX) and cold (CHX) heat exchangers respectively.

In reality we do not use a mechanical piston, but a complex device called a compressor, which produces a pressure wave [29],

$$p(t) = p_0 + p_1 \cos(\omega t).$$

where p_0 is the mean pressure of the environment, p_1 is the pressure amplitude, and ω is the angular frequency of oscillation. The pressure wave drives mass flow of the gas, as a pressure gradient is created. Once the gas reaches the end of the PTC, it travels down a new component the delay mechanism. Whose purpose is to create a phase difference between the mass flow rate $\dot{m}$ and p,

$$\dot{m} = \dot{m}_0 \cos(\omega t - \phi).$$

The reader can think of it like how water waves move sand underneath it, there is a visual delay in the movement of the sand due to the sand's inertia, compared to the peaks in the water wave. The gas particles have inertia when going down the inertance tube which is designed to resist rapid acceleration. Hence the gas particles lag behind the pressure wave when coming back out of the delay mechanism once reflected back.

The manufacturer has then placed the CHX and HHX and timed the compressors oscillations perfectly so that when gas arrives at the CHX, the pressure wave their is entering the expansion phase. The gas expands becoming colder than the CHX and hence heat flows from CHX into the gas. And similarly when the gas arrives at the HHX, the pressure wave is entering its compression phase. Temperature rises by a considerable amount, and heat flows from the hot gas to the colder HHX, venting the heat out.

Upon the gas moving back from the HHX to CHX we introduce another new component the regenerator to help efficiently cool the HHX gas going towards the CHX [30]. The regenerator does this by creating its own temperature gradient from the CHX to the HHX using a structure that conducts heat well between a gas and solid say a porous high heat capacity material a thermal sponge, but doesn't conduct heat length-wise disconnecting the heat that would flow from the HHX to the CHX, this can be done by having gaps between the thermal sponge. This temperature gradient means as the gas flows from the CHX which is now hot, it starts to cool down as it transfers heat to the CHX side of the regenerator, then as the gas moves along to the HHX side of the regenerator the gas transfers less heat. More importantly after the gas has gone through its inevitable compression stage and vented as much heat from this process out at the HHX, the gas travels back down the regenerator. Now the temperature gradient of the regenerator really helps to cool the gas that is moving back towards the CHX. Making the expansion stage at the CHX more efficient. See Image 6 for how all the components are laid out.

The chosen gas for the PTC is ^4He [30]. This is because it remains a gas even at extremely low temperatures. Has a low viscosity so less heat generated by friction during oscillations. Has a high thermal conductivity allowing for efficient heat transfer between the gas and components. And is a noble gas so does not react chemically with other materials in the PTC, hence does not erode components increasing the longevity of the PTC.

If we chain two PTCs together. We can use the first stage to cool from $300K \rightarrow 30K$, and then this stage can pre-cool the second stage which now can cool from $30K \rightarrow 4K$. The coldest part of the second stage, the CHX is our copper plate that is thermally linked to the next component of the cryostat. The 2-stage PTC described above was what the research group at the University of Manchester that our project was attached to used. In particular the SHI Cryogenics RP-082B2 [31], from now on we will continue to refer to this 2-stage PTC as a PTC.

3.2 $^{3/4}He$ Sorption Cooler ($4K \rightarrow 0.3K$)

The second cooling stage in the cryostat employs a sorption cooler to achieve temperatures from $4K$ down to approximately $300mK$. Sorption cooling operates through the combined effects of evaporation and adsorption in helium systems.

For a single stage ($^{3/4}$He) sorption cooler, a liquid helium pot thermally connected to the $4K$ flange of the PTC, with a cryopump containing activated charcoal positioned above it, see Figure 8a. Activated charcoal is used due to its very large specific surface area $> 1000m^2g^{-1}$ [33], enabling

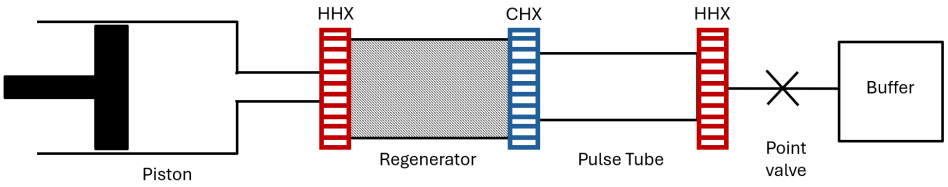


Figure 6: Schematic diagram of a PTC and all it's components. Adapted from [32].

efficient adsorption of He gas atoms via van der Waals interactions (physisorption) at the surface of the charcoal pores.

Owing to its highly porous structure, see Image 7, adsorption occurs not only at the external surface of the material but predominantly within internal pores, providing a large number of available binding sites. The high specific surface area reflects the abundance of these internal pore surfaces, increasing the number of adsorption sites available to helium atoms [33]. This allows helium gas to permeate the charcoal structure while remaining bound to pore surfaces, rather than being absorbed into the bulk material.

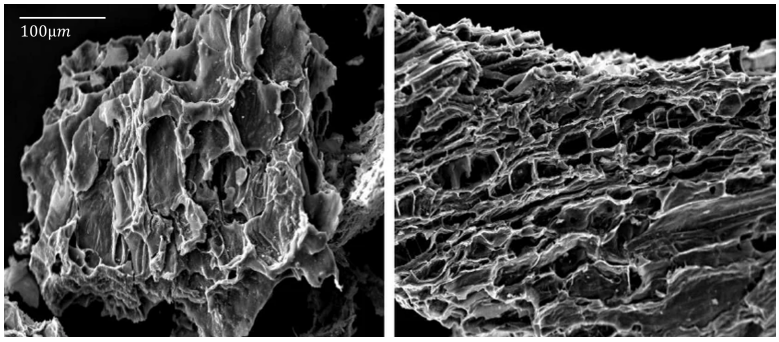


Figure 7: Activated Charcoal pores taken from a Scanning Electron Microscope (SEM) [34].

At $T \sim 4K$ the thermal energy of the He gas atoms cannot overcome the van der Waals forces at the surface of the charcoal (adsorption phase), but at $T \sim 50K$ the atoms do have enough energy to separate (desorption phase) [33].

The cooling mechanism relies on reducing the vapour pressure, P_{vap}, above the liquid helium. For a liquid–vapour system in equilibrium (co-existence curve on a $P-T$ diagram), the vapour pressure is given by the saturated vapour pressure, P_{sat}, which is the unique equilibrium pressure at which liquid and vapour coexist at temperature T ($P_{vap} = P_{\text{sat}}$). P_{sat} follows the Clausius–Clapeyron relation [35],

$$\frac{dP_{\text{sat}}}{dT} = \frac{L}{T\Delta V} \approx \frac{LP}{RT^2} \implies P_{\text{sat}}(T) \propto \exp\left(-\frac{L}{RT}\right), \tag{19}$$

where L is the latent heat of vaporisation and R is the molar gas constant.

When the charcoal is cooled (by thermal contact with the $4K$ plate), it adsorbs helium vapour, lowering the vapour pressure such that $P_{vap} < P_{\text{sat}}(T)$. This drives evaporation of the liquid helium to restore equilibrium ($P_{\text{sat}} \to P_{vap}$). And from equation (19), $P_{\text{sat}}(T) \downarrow \Rightarrow e^{-1/T} \downarrow \Rightarrow T \downarrow$, hence cooling.

The cooling power of the system (rate of heat removal) is then determined by the rate of liquid helium evaporation. This can be written as

$$\dot{Q} = \dot{n}_l \left(H_{\text{vapour}} - H_{\text{liquid}}\right) = \dot{n}_l L, \tag{20}$$

where $\dot{n}_l$ is the molar evaporation rate of liquid helium (mols^{-1}) and H_{phase} is the molar enthalpy of each respective phase (J mol^{-1}) [36]. This evaporation extracts latent heat from the liquid and the attached cold stage, thereby reducing its temperature.

The evaporation rate is limited by the total amount of liquid helium available and the operating time Δt, such that

$$\dot{n}_l = \frac{n_l}{\Delta t}. \tag{21}$$

As a result, sorption coolers are inherently finite-duration (single-shot) systems, with typical hold times on the order of several hours.

Not all helium within the system contributes directly to cooling. The total number of moles can be decomposed as

$$n_{\text{tot}} = n_l + n_e + n_p + n_t, \tag{22}$$

where n_l is the liquid helium in the evaporator, and the remaining terms correspond to vapour in the evaporator n_e, cryopump n_p, and connecting tubing n_t respectively [33]. Only n_l contributes to cooling power, highlighting the importance of efficient condensation and minimisation of vapour losses.

In addition to the finite helium supply, the cooling power is limited by parasitic thermal loads, the most dominant being conduction through the stainless steel tubing. This heat leak can be expressed as

$$\dot{Q}_{\text{cond}} = \frac{A}{l} \int_{T_{\text{cold}}}^{T_{\text{hot}}} \kappa(T)\, dT, \tag{23}$$

where $\kappa(T)$ is the temperature-dependent thermal conductivity of the stainless steel [33]. As the helium is depleted, the evaporation rate decreases ($\dot{n}_l \to 0$), and hence the cooling power vanishes ($\dot{Q} \to 0$), marking the end of the cooling cycle.

To reset the cycle, the charcoal is heated to $T \sim 50K$, causing desorption of helium atoms. This increases the vapour pressure above the liquid, allowing the gas to recondense into the pot. The system therefore operates as a single-shot refrigerator, limited by helium depletion or adsorber saturation.

A two-stage configuration is used to reach sub-Kelvin temperatures. The first stage employs ^4He, which condenses at $\sim 4.2K$ and can cool to approximately $T \approx 1K$. The second stage uses ^3He, which has a lower boiling point ($3.19K$ at 1 atm) and therefore requires a colder condensation surface. This is achieved by thermally linking the ^3He condensation line to the coldest part of the ^4He stage, the ^4He pot.

In a two-stage system, the cooling power of the ^4He stage must also support the thermal load of the ^3He stage, such that

$$\dot{Q}_{^4He} = \dot{Q}_{^3He} + \dot{Q}_{\text{parasitic}} \approx \dot{Q}_{^3He} + \dot{Q}_{\text{cond}}.$$

This coupling places additional constraints on the design, as the ^4He stage must provide sufficient cooling to both maintain its own temperature and enable condensation of ^3He. The cooling power of each stage can be written as

$$\dot{Q}_i = \dot{n}_{l_i} L_i, \quad i \in \{^3He, {}^4He\}, \tag{24}$$

with ^3He providing the final stage of cooling. By pre-cooling the ^3He using the ^4He stage, the system can reach temperatures as low as $\sim 300mK$.

This combined ^3He–^4He configuration, commonly referred to as a "He-7" sorption cooler, typically operates for several hours before requiring recycling.

3.2.1 Volume Estimation

As an example of a calculation using equations from Section 3.2. We can calculate the amount of $^{3/4}He$ needed to operate the "He-7" sorption cooler for $\Delta t = 5$ hours.

As we will be utilising the "He-7" sorption cooler during it's steady state, ie. when the temperature is constant. This means no net heat flow. So the cooling power (20) exactly balances the heating (23),

$$\dot{Q} = \dot{Q}_{\text{parasitic}} \approx \dot{Q}_{\text{cond}} \implies \dot{n}_l L = \dot{Q}_{\text{cond}}$$

where $L_{^3He} \approx 25 Jmol^{-1}$ and $L_{^4He} \approx 85 Jmol^{-1}$ [33]. If $\dot{n}_l L > \dot{Q}_{\text{cond}}$ you're removing more heat than is coming in, the system cools further. And if $\dot{n}_l L < \dot{Q}_{\text{cond}}$ then heat leaks dominate, temperature rises. Only when they're equal do you get a stable lowest temperature.

To calculate Q_{cond} for each stage, requires $\kappa(T)$ for Stainless steel. For our setup we are using 304 Stainless steel, and one can find the equation of the line that describes the thermal conductivity against temperature from the National Institute of Standards and Technology (NIST) [27], see Figure 8b.

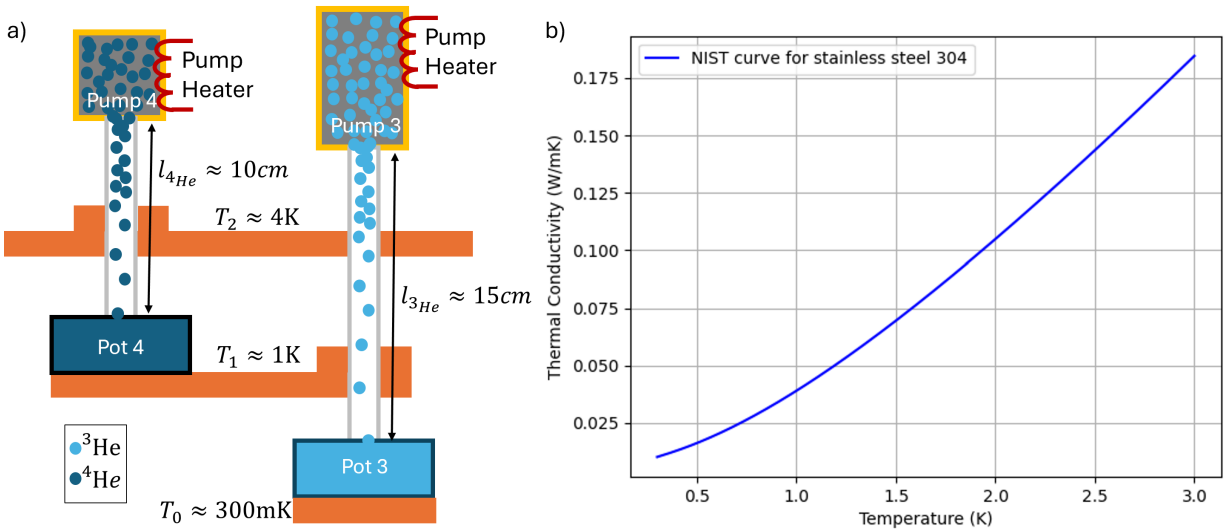


Figure 8: a) Dimensions of a two-stage sorption cooler with measurements for the Volume estimation Section 3.2.1. Adapted from [33], b) Thermal conductivity of Stainless steel 304 from $0.3K$ to $3K$ taken from NIST data [27].

Using this data and the dimensions from Figure 8a we can calculate Q_{cond}, using equation (23), for the stainless steel tube for both $^{3/4}He$ sorption coolers using the Numpy Trapezoid function [37],

$$\text{For } {}^4He \implies \dot{Q}_{\text{cond}} = \frac{A}{l_{^4He}} \int_{T_1=1K}^{T_2=4K} \kappa(T)\, dT \approx 29.9\mu W,$$

$$\text{For } {}^3He \implies \dot{Q}_{\text{cond}} = \frac{A}{l_{^3He}} \int_{T_0=0.3K}^{T_1=1K} \kappa(T)\, dT \approx 0.722\mu W,$$

where A is the area of the cross section of the stainless steel tube with thickness $\sim 22mm$, a annulus.

Now to calculate n_l we simply use equation (21) and (20), but this isn't the total amount of helium within the system as is pointed out by equation (22). If we assume ideal gas we can calculate the other components as follows [33]:-

- In evaporator:
$$n_e = \frac{P(V_e - n_l \tilde{V})}{RT_e},$$
where $\tilde{V}$ is molar volume of liquid ^4He at $1K$.
- In Stainless Steel(SS) tube:
$$n_t = \int_0^l P \frac{\pi d^2}{4} \frac{dz}{T(z)},$$
where temperature of the SS tube, $T(z)$, varies linearly from T_p to T_e.
- In cryopump:
$$n_p = \frac{PV_p}{RT_p}$$
where the Volumes of cryopump (V_p) and evaporator (V_e) are assumed to be $\sim 20cm^3$

Calculating these values. And subbing into equation (22) along with n_l gets the total amounts of Helium required to operate the sorption cooler for $\Delta t = 5$ hours. See Table 1 for results of the estimation. It is key to remember this is an estimation. And if designing your own sorption cooler you should take these results as a lower bound for how much $^{3/4}He$ you will need.

He	Total Moles (mol)	Required volume at STP (L)
4	0.149	3.38
3	0.303	6.88

Table 1: Results of estimation showing required volume at Standard Temperature and Pressure (STP) to operate sorption coolers for 5 hours

3.3 Split Condenser

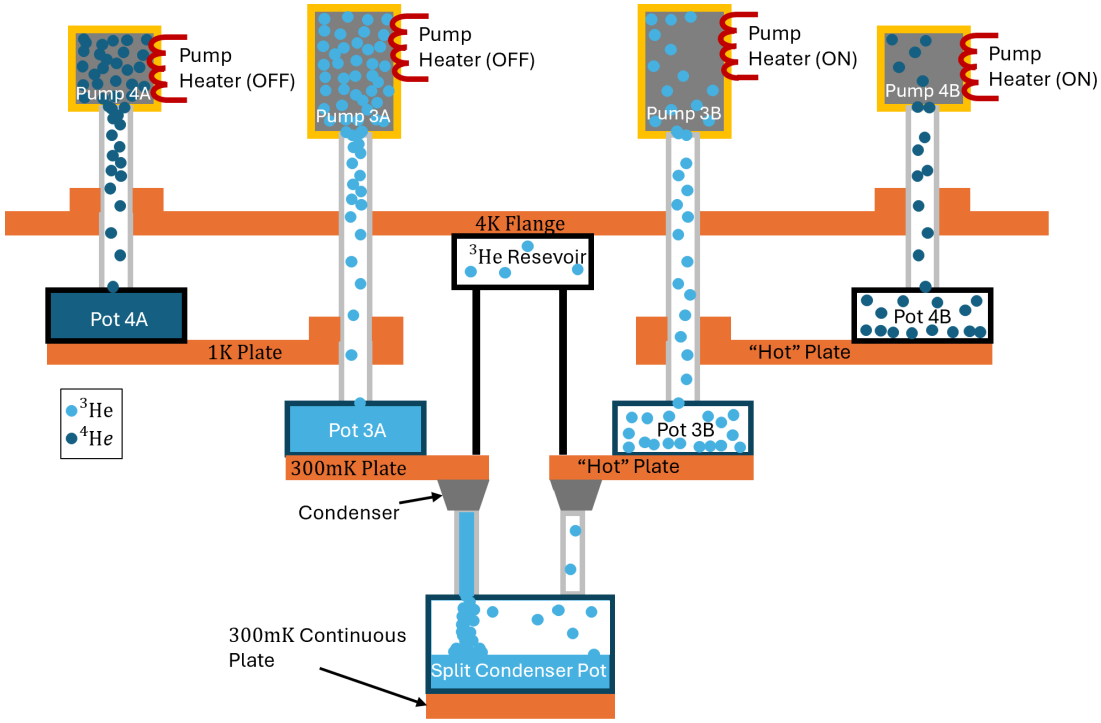


Figure 9: Schematic diagram of "He-7" sorption cooler working in anti-phase, with side A in cooling phase and side B in regeneration phase. Adapted from [38].

The "He-7" sorption cooler discussed in Section 3.2 and the volume estimation in Section 3.2.1 operates as a single-shot refrigerator with a hold time of approximately 5 hours. Extending the operating time of a single stage for longer hours would require more $^{3/4}He$. However, by mirroring the "He-7" sorption cooler configuration, shown schematically in Figure 9 as side A and side B, and operating the two systems in anti-phase, continuous cooling near $300mK$ can be achieved indefinitely [39]. In this configuration, one unit provides active cooling while the other undergoes regeneration, during which the cryopump heater is activated.

To thermally isolate the regenerating stage from the actively cooling stage, the system employs a split condenser. The split condenser is thermally linked to the coldest point of each ^3He sorption stage and acts as an intermediate condensation unit. During operation, the actively cooling sorption stage maintains its condenser at approximately $300mK$, well below the $3.19K$ boiling point of ^3He, causing helium gas to condense into liquid form. The condensed liquid helium then flows into the split condenser pot under gravity [39].

On the regenerating side, although the cryopump heater reaches temperatures of $\sim 50K$, the circulating helium gas is partially cooled through thermal contact with the continuously operating $4K$ stage. Consequently, helium atoms can still migrate toward the colder condenser where liquefaction is energetically favourable. This establishes a continuous circulation process driven by the temperature gradient between the warm and cold sides of the system, analogous to a convection current. Heat is therefore continuously removed from the split condenser pot and transported through the helium circulation loop, allowing the cold stage temperature to remain near $300mK$ despite periodic regeneration cycles [39].

The operation of the split condenser can be experimentally verified using temperature data obtained from the cryostat used by the research group at the University of Manchester that our project was associated with. Figure 10 shows the temperatures of Pot 3A, Pot 3B, and the split condenser pot during anti-phase cycling.

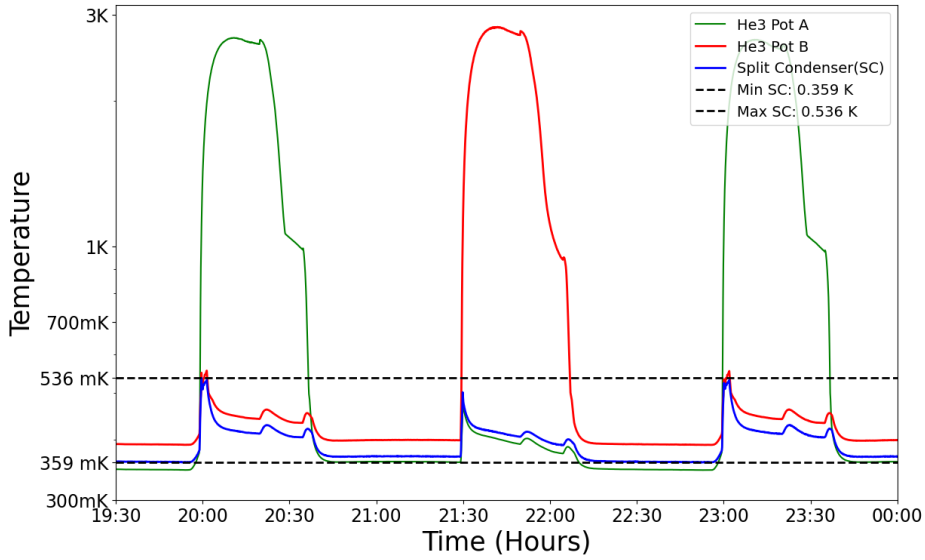


Figure 10: Temperature of Pot 3A, Pot 3B, and the split condenser pot during anti-phase cycling. Temperature from the cryostat of the research group at the University of Manchester that our project was associated with.

As one sorption stage undergoes regeneration and increases in temperature, the second stage remains in its cooling cycle near $300mK$. Despite the periodic regeneration cycles causing the split condenser temperature to fluctuate between approximately $359mK$ and $536mK$, the condenser consistently remains well below 1K and preferentially couples to the actively cooling side of the system. This behaviour confirms the continuous cooling mechanism described above.

Finally, a ^3He expansion reservoir is incorporated into the system to accommodate pressure variations, particularly at elevated temperatures [39], ensuring that

$$\Delta P = \frac{nRT}{V}$$

remains within safe operational limits.

3.4 Miniature Dilution Refrigerator ($0.3K \rightarrow 0.05K$)

A miniature dilution refrigerator (mini-DR) constitutes the final cooling stage in our cryogenic system and operates using a mixture of helium isotopes, $^{3/4}He$. Its operation relies on the unique low-temperature thermodynamic properties of this mixture, particularly the phase separation.

To initiate operation, the system must first be pre-cooled to temperatures below $T \lesssim 0.85K$, via our sorption cooling stage which cools to $\sim 300mK$ [33]. At low temperatures, ^4He, a boson, undergoes a transition below its lambda point ($T_\lambda \approx 2.17K$ [36]) into a superfluid phase described by Bose-Einstein statistics. In contrast, ^3He, a fermion, remains a normal Fermi liquid governed by the Pauli exclusion principle down to millikelvin temperatures (its superfluid transition occurring at $\sim 0.94mK$, far below our mini-DR's operating regimes).

Above $\sim 0.85K$, the $^{3/4}He$ mixture is homogeneous. However, below this temperature, the system enters a phase-separated regime [29], forming two distinct phases:-

- Dilute phase, consisting primarily of superfluid ^4He with a small fraction ($\sim 6.6\%$ as $T \rightarrow 0$) of dissolved ^3He.

- Concentrated phase, rich in ^3He with some ^4He.

See Figure 11 for the Phase diagram of $^{3/4}He$.

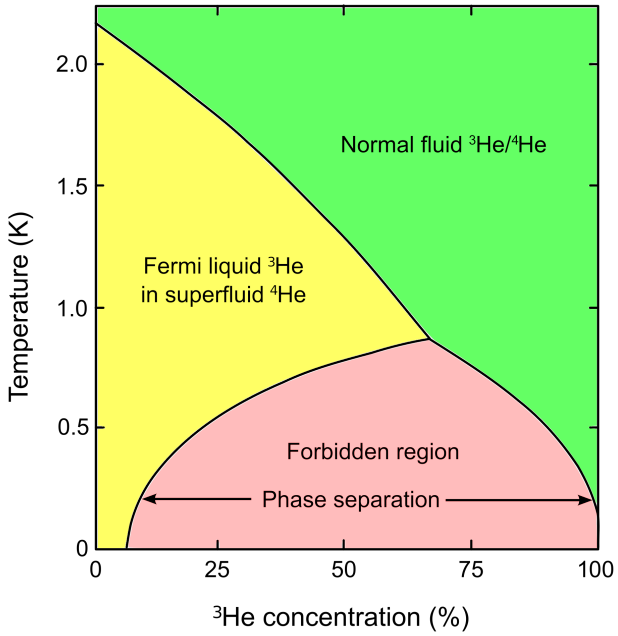


Figure 11: Phase diagram of $^{3/4}He$ mixture with Temperature (K) against concentration (%) of ^3He. The curves separate the phases of $^{3/4}He$ mixture into Dilute (Fermi liquid ^3He in Superfluid ^4He) and Concentrated (Normal fluid $^{3/4}He$) phase. Taken from [40].

This phase separation of mixtures is like oil in water, they don't mix, and have a physical boundary between the constituent parts. And in the ideal stable case, the 2 phases will stratify, due to gravity. As the concentrated phase is lighter than the dilute phase it will sit above the dilute phase in the component of the mini-DR known as the mixing chamber (MC). The $^{3/4}He$ mixtures in the MC can be described by the 1st law of thermodynamics,

$$dU = TdS - PdV + \mu_c dN_c + \mu_d dN_d, \tag{25}$$

where $U = U(S, V, N)$ is the internal energy of the system, T, S, P and V is the temperature, entropy, pressure and volume of the system, respectively and μ is the chemical potential, defined such that $\mu_{phase} dN_{phase}$ is the energy change from adding particles N to each phase. If we make a Legendre transformation, $U(S, V, N) \to G(T = \frac{\partial U}{\partial S}, P = -\frac{\partial U}{\partial V}, N) = \underbrace{U + PV}_{H} - TS$ to get the Gibbs free energy, we get

$$dG = -SdT + VdP + \mu_c dN_c + \mu_d dN_d.$$

As in our ideal stable case, when the phase boundary is stable, the MC can be assumed to be at constant T and P. The total number of $^{3/4}He$ in the MC remains constant, $N_{tot} = N_c + N_d \Rightarrow dN_d = -dN_c$,

$$\implies dG = (\mu_c - \mu_d) dN_c. \tag{26}$$

If we use the 2nd law of thermodynamics which implies that for an isolated system in equilibrium, the total entropy is maximised, $dS_{tot} = 0$ (& $\delta S^2_{tot} < 0$). The total entropy for our "closed" system is $S_{tot} \approx S_{\text{mixture}} + S_{\text{MC wall}}$, and using (25) this means,

$$dS_{tot} = dS_{\text{mixture}} + dS_{\text{MC wall}},$$

$$TdS_{tot} = (dU + PdV - \mu_c dN_c - \mu_d dN_d) + (-dU - PdV) = -dG,$$

$$\implies dG = 0 \quad (\Delta G = 0), \tag{27}$$

where $dS_{\text{MC wall}} = -dU - PdV$, because when the $^{3/4}He$ mixtures gain energy dU, the MC walls lose that same energy, and if the $^{3/4}He$ mixtures expands by dV, the MC walls do work. So using this result and equation (26), and the fact that dN_c is a free variable as the system must be stable against any infinitesimal particle exchange, this means,

$$\mu_c = \mu_d.$$

This chemical potential equilibrium can be disturbed by changing the concentration of ^3He in the dilute phase. Removing some ^3He from the dilute phase will disturb this equilibrium ($\mu_c > \mu_d$), such that at $T \to 0K$, the concentration of ^3He in the dilute phase $X_{3D} < 6.6\%$. To restore equilibrium, ^3He from the concentrated phase will move to the dilute phase, it does this because the binding energy of $^3He - ^4He$ bonds in the dilute phase is greater than the binding energy of $^3He - ^3He$ bonds in the concentrated phase [33]. So its energetically favourable for ^3He to move to the dilute phase. The act of moving across this boundary, absorbs heat from the surroundings (i.e. the MC and attached load). To calculate the cooling power for this process we use the definition of Enthalpy and (27) to get,

$$\Delta G = \Delta H - T\Delta S \implies \Delta H = T\Delta S$$

$$H_D - H_C = T_{\text{MC}}(S_D - S_C) = T_{\text{MC}}(95T_{\text{MC}} - 11T_{\text{MC}}) = 84T_{\text{MC}}^2$$

where $S_D = 95T\ Jmol^{-1}K^{-1}$ and $S_C = 11T\ Jmol^{-1}K^{-1}$ are empirically found values from [29]. Using the same equation (20) from Section 3.2 we get the cooling power,

$$\dot{Q} = 84\dot{n}_3 T_{\text{MC}}^2 \implies T_{\text{MC}} = \sqrt{\frac{\dot{Q}}{84\dot{n}_3}}. \tag{28}$$

Which shows that to reach lower temperatures we must reduce the heat load $\dot{Q}$ and increase the circulation rate of ^3He $\dot{n}_3$. The rate at which ^3He crosses the phase boundary.

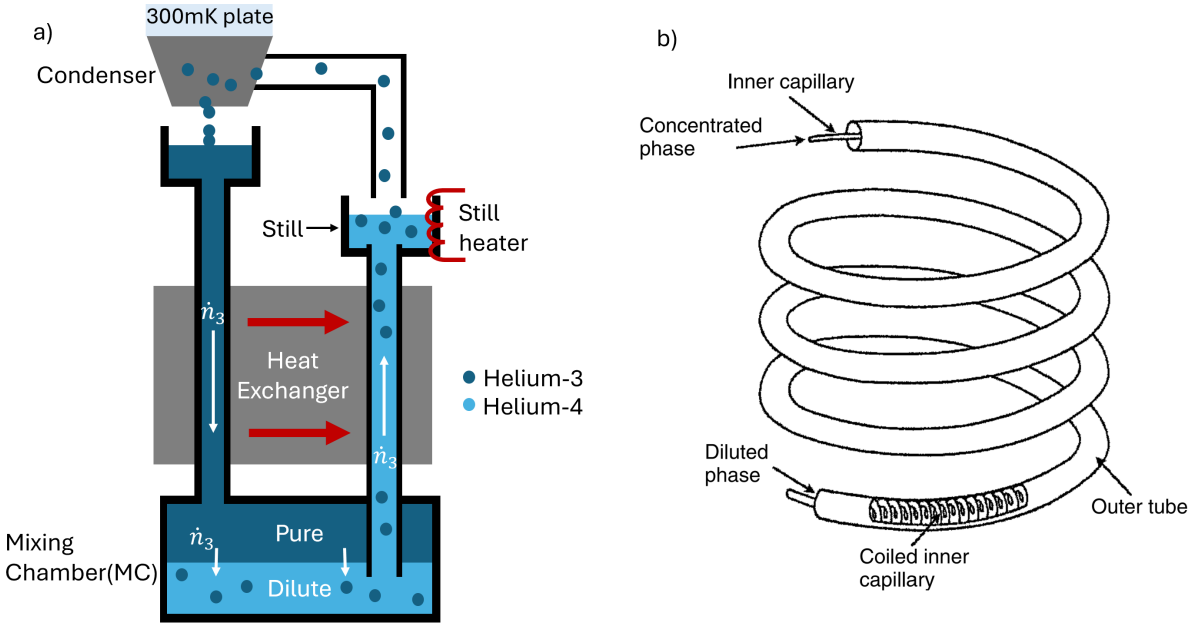


Figure 12: a) Schematic diagram of a mini-DR, adapted from [29], b) Schematic diagram of counterflow heat exchanger [29].

To sustain continuous cooling, this process requires continuous removing of ^3He from the dilute phase, so that ^3He continues to cross the phase boundary from the concentrated phase. This is done via a still (from the word "distillation"), as you can see in Image 12a, which preferentially evaporates ^3He rather than ^4He. The preferential evaporation process works via a similar mechanism, the Clausius-Claperyon relation, as described in Section 3.2, where a liquid-vapour system in equilibrium means the

vapour pressure above the still is equal to the saturated vapour pressure at the still ($P^i_{\text{above}} = P^i_{\text{sat}}(T_{\text{still}})$ for each component of the mixture $i = {}^3He, {}^4He$) [35]. Now the heater at the still ($T_{\text{still}} \approx 0.7K$) starts the process of evaporation, and as the gas travels up to the $300mK$ condenser plate, the gas condenses reducing the pressure above the still ($P^i_{\text{above}} < P^i_{\text{sat}}(T_{\text{still}})$). So to restore the equilibrium pressure the system evaporates more of the ${}^{3/4}He$ dilute mixture increasing the P^i_{above}, such that $P^i_{\text{above}} \to P^i_{\text{sat}}(T_{\text{still}})$. The preferential aspect comes from the fact that $P^3_{\text{sat}}(T_{\text{still}}) >> P^4_{\text{sat}}(T_{\text{still}})$, due to equation (19) and the latent heats being empirically different $L_3 < L_4$. This means ${}^3He$ is much further from equilibrium and hence evaporates much faster.

The reader maybe wondering how ${}^3He$ is able to overcome gravity and move towards the still which sits above the MC. This is best explained via Van't Hoff equation which states that a substance dissolved in a fluid medium behaves as if it were in a vacuum, and hence behaves as an ideal gas [33]. In our situation ${}^4He$ behaves as our vacuum, and ${}^3He$ obeys Van't Hoff's equation,

$$\Pi V_{m,4} = X_{3D} RT,$$

which looks very similar to the ideal gas law but where Π is the osmotic pressure and $V_{m,4} = V/n_4$ is the molar volume of the solvent ${}^4He$. We can therefore write the osmotic pressure difference between the still and MC as,

$$\Delta\Pi = \Pi_{\text{MC}} - \Pi_{\text{still}} = \frac{(X_{\text{MC}} T_{\text{MC}} - X_{\text{still}} T_{\text{still}})R}{V_{m,4}}$$

where $X_{\text{MC/still}}$ is the concentration of ${}^3He$ in the MC and still respectively. As ${}^3He$ is evaporated from the still, $X_{\text{still}} << X_{\text{MC}} \approx 6.6\%$, this creates a non-negative osmotic pressure difference $\Delta\Pi \neq 0$, which drives the ${}^3He$ freely up to the still.

The last component of the mini-DR not mentioned that can be seen in Figure 12a is the counterflow heat exchanger [29]. This is an efficiency tool, best described as contact cooler where heat from the incoming "hot" concentrated phase ${}^3He(300mK)$ that is travelling towards the MC comes into contact with the outgoing colder dilute phase ${}^{3/4}He(50mK)$ travelling to the still.

In Figure 12a it isn't obvious how this is done, but if you look at Figure 12b, you can see how this is achieved by an outer tube containing the colder outgoing dilute phase cooling by contact the inner tube containing the hotter concentrated phase. We can use the same cooling power equation from before (28), which was an idealised case that assumed the incoming ${}^3He$ was at T_{MC}, but if we add the parasitic heat load from the incoming ${}^3He$ that is at the temperature of the heat exchanger T_{ex} we get,

$$\dot{n}_3\big(H_{3D}(T_{\text{MC}}) - H_{3C}(T_{\text{MC}})\big) = \dot{Q} + \underbrace{\dot{n}_3\big(H_{3C}(T_{\text{ex}}) - H_{3C}(T_{\text{MC}})\big)}_{\text{parasitic heat load}} \tag{29}$$

where $H_{3D}(T) \approx H_3(0) + 95T^2$ and $H_{3C}(T) \approx H_3(0) + 11T^2$ from [29]. Hence equation (29) becomes,

$$84\dot{n}_3 T^2_{\text{MC}} = \dot{Q} + \dot{n}_3\left(11T^2_{\text{ex}} - 11T^2_{\text{MC}}\right)$$

$$\implies \dot{Q} = \dot{n}_3\left(95T^2_{\text{MC}} - 11T^2_{\text{ex}}\right)$$

If the heat exchanger was perfect $T_{\text{ex}} = T_{\text{MC}}$ we would recover equation (28). Now for cooling to occur we need $\dot{Q} \geq 0$, as this shows that heat is being absorbed from the surroundings,

$$95T^2_{\text{MC}} - 11T^2_{\text{ex}} \geq 0$$

$$\implies T_{\text{ex}} \lesssim 3T_{\text{MC}}$$

This means that if the $T_{\text{ex}} > 3T_{\text{MC}}$ the refrigerator will stop cooling entirely. So the heat exchangers efficiency is important.

The operation of the miniature dilution refrigerator can be experimentally verified using temperature data obtained from the cryostat used by the research group at the University of Manchester that this project was attached to. Figure 13 shows the temperatures of the split condenser stage and the mixing chamber during operation of the mini-DR.

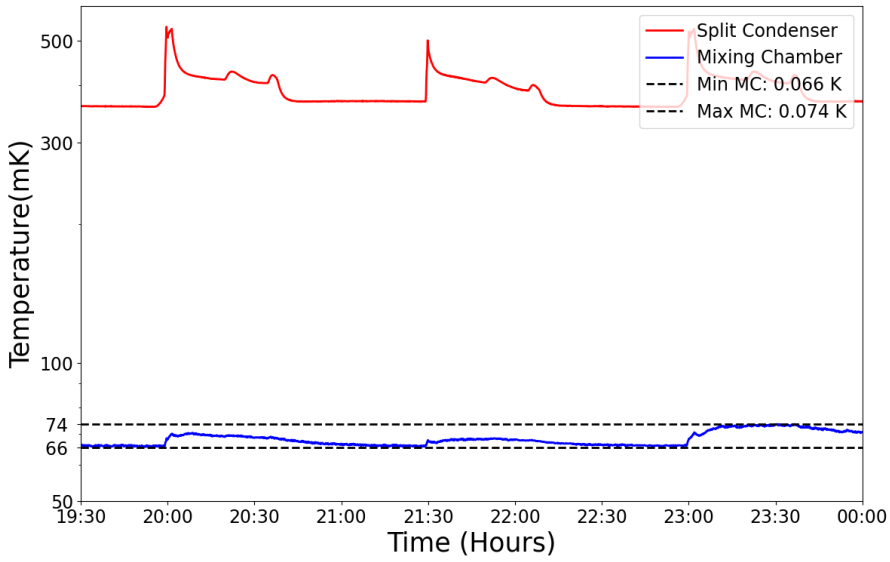


Figure 13: Temperature of split condenser and mixing chamber of mini-DR. Temperature from the cryostat of the research group at the University of Manchester that our project was associated with.

As the split condenser remains at temperatures where the $^{3/4}He$ mixtures have a phase separation, the dilution refrigerator can therefore further cool the mixing chamber through the continuous transfer of ^{3}He atoms from the concentrated phase into the dilute phase. The measured mixing chamber temperature fluctuates between approximately $66mK$ and $74mK$, shown by the horizontal bounds in Figure 13.

Although this is slightly higher than the desired operating temperature of $\sim 50mK$ discussed throughout this report, it still demonstrates successful cooling well into the millikelvin regime. The difference is likely due to residual parasitic heat leaks, imperfect heat exchange, and non-ideal ^{3}He circulation.

Overall, the essential cooling mechanism of a dilution refrigerator is the continuous, transfer of ^{3}He atoms from a concentrated phase into a dilute superfluid phase, enabled by phase separation and maintained through selective evaporation and recirculation, boosted in performance by a heat exchanger.

4 Conclusion

This report has examined the physical requirements for operating a compact superconducting quantum computer based on transmon qubits. Starting from basic quantum information theory, it was shown that a useful qubit requires an anharmonic energy spectrum so that the computational subspace $\{|0\rangle, |1\rangle\}$ can be selectively addressed without strongly populating higher excited states. Josephson junction circuits provide this required non-linearity through the cosine potential, and the transmon regime $E_J/E_C \gg 1$ suppresses charge sensitivity while retaining sufficient anharmonicity for qubit operation.

The operational requirement for cryogenic cooling was then motivated using the typical transmon transition frequency $f_{01} \sim 6\,\text{GHz}$. This corresponds to a thermal energy scale of approximately $300mK$, meaning that operation well below this temperature is required to suppress thermal photon occupation and maintain reliable ground-state initialization. This motivates a target operating temperature of order $\sim 50mK$ for superconducting quantum computation.

To achieve this, the cryostat was analysed as a sequence of cooling stages. The PTC provides the initial cooling from room temperature to approximately $4K$, enabling the operation of the He sorption cooler. The $^{3/4}He$ sorption system then cools to the sub-kelvin regime, with the split condenser

allowing continuous operation by cycling two sorption stages in anti-phase. Experimental temperature data showed that the split condenser remained below $1K$ during regeneration cycles, confirming that the system successfully provides the required pre-cooling for the mini-DR.

The final stage, the mini-DR, uses the phase separation of a $^{3/4}He$ mixture to cool the mixing chamber through the continuous transfer of ^{3}He atoms from the concentrated phase into the dilute phase. The measured mixing chamber temperature fluctuated between approximately $66mK$ and $74mK$. Although this is slightly above the desired $\sim 50mK$ operating temperature discussed in this report, it still demonstrates successful cooling well into the millikelvin regime and close to the required temperature scale for superconducting qubit operation. The remaining difference is likely due to residual parasitic heat leaks, imperfect counterflow heat exchange, and non-ideal ^{3}He circulation.

Further work could improve the connection between the cryogenic performance and qubit operation. In particular, a more complete treatment of the resonator and transmission-line environment, for example through the Jaynes-Cummings Hamiltonian, would allow the readout and control architecture of the transmon to be discussed in more detail. Experimentally, analysing temperature data from the still would allow the ^{3}He circulation rate $\dot{n}_3$ to be estimated and compared with the cooling-power relation (28)

$$\dot{Q} = 84\dot{n}_3 T_{\mathrm{MC}}^2.$$

This would make it possible to determine whether the mixing chamber temperature is limited primarily by insufficient circulation rate or by excess heat load. Additional temperature sensors placed at key locations, such as the heat exchanger, condenser, support structures, and wiring stages, would also help identify dominant heat leaks. Finally, although not central to this project, future work could also consider the role the heat switches played and whether their thermal isolation during regeneration was the most efficient for the cooling cycle.

Overall, the project shows that compact cryogenic systems based on PTC, sorption refrigeration, and mini-DR can reach the millikelvin regime required for superconducting quantum computing. The measured performance demonstrates that the cryostat is already close to the target operating temperature, and further optimisation of heat loads, heat exchange, and ^{3}He circulation could allow the system to approach the desired $\sim 50mK$ regime more reliably.

References

[1] Peter W. Shor. "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer". In: *SIAM Journal on Computing* 26.5 (Oct. 1997), pp. 1484–1509. ISSN: 1095-7111. DOI: 10.1137/s0097539795293172. URL: http://dx.doi.org/10.1137/S0097539795293172.

[2] R. L. Rivest, A. Shamir, and L. Adleman. "A method for obtaining digital signatures and public-key cryptosystems". In: *Commun. ACM* 21.2 (Feb. 1978), pp. 120–126. ISSN: 0001-0782. DOI: 10.1145/359340.359342. URL: https://doi.org/10.1145/359340.359342.

[3] Charles H. Bennett and Gilles Brassard. "Quantum cryptography: Public key distribution and coin tossing". In: *Theoretical Computer Science* 560 (Dec. 2014), pp. 7–11. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2014.05.025. URL: http://dx.doi.org/10.1016/j.tcs.2014.05.025.

[4] Sheng-Kai Liao, Wen-Qi Cai, Wei-Yue Liu, et al. "Satellite-to-ground quantum key distribution". In: *Nature* 549.7670 (2017), pp. 43–47. DOI: 10.1038/nature23655.

[5] Yu-Ao Chen, Qiang Zhang, Teng-Yun Chen, et al. "An integrated space-to-ground quantum communication network over 4,600 kilometres". In: *Nature* 589 (2021), pp. 214–219. DOI: 10.1038/s41586-020-03093-8.

[6] IBM Quantum. *IBM Quantum brings utility scale to the world.* Accessed: 14 May 2026. 2023. URL: https://www.ibm.com/quantum/blog/ibm-quantum-roadmap-2025.

[7] John Preskill. "Quantum Computing in the NISQ era and beyond". In: *Quantum* 2 (2018), p. 79. DOI: 10.22331/q-2018-08-06-79.

[8] Jens Koch, Terri M. Yu, Jay Gambetta, et al. "Charge-insensitive qubit design derived from the Cooper pair box". In: *Physical Review A* 76.4 (2007), p. 042319. DOI: 10.1103/PhysRevA.76.042319.

[9] All About Circuits. *Critical Temperatures for Superconductors*. Accessed: 3rd March 2026. 2026. URL: https://www.allaboutcircuits.com/textbook/reference/chpt-3/critical-temperatures-for-superconductors/.

[10] John Bardeen, Leon N. Cooper, and J. Robert Schrieffer. "Theory of Superconductivity". In: *Physical Review* 108.5 (1957), pp. 1175–1204. DOI: 10.1103/PhysRev.108.1175.

[11] Antonio Vettoliere, Paolo Silvestrini, and Carmine Granata. "3 - Superconducting quantum magnetic sensing". In: (2023). Ed. by Mohamed Henini and Marcelo Oliveira Rodrigues, pp. 43–85. DOI: https://doi.org/10.1016/B978-0-12-820566-2.00001-6. URL: https://www.sciencedirect.com/science/article/pii/B9780128205662000016.

[12] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Josephson Junction*. The Feynman Lectures on Physics, Volume III, Chapter 21, Section 21-9. Accessed: 14 May 2026. URL: https://www.feynmanlectures.caltech.edu/III_21.html.

[13] Phillip M. Duxbury. *Ginzburg-Landau Theory, Modeling of Dynamics and Fluctuations*. Michigan State University PHY 831 Lecture Notes. Accessed: 14 May 2026. URL: https://web.pa.msu.edu/people/duxbury/courses/phy831/LectureNotesAndProblemsPart4.pdf.

[14] V. L. Ginzburg and L. D. Landau. "On the Theory of Superconductivity". In: (2009). DOI: 10.1007/978-3-540-68008-6_4. URL: https://doi.org/10.1007/978-3-540-68008-6_4.

[15] S.E. Rasmussen et al. "Superconducting Circuit Companion—an Introduction with Worked Examples". In: *PRX Quantum* 2 (4 Dec. 2021), p. 040204. DOI: 10.1103/PRXQuantum.2.040204. URL: https://link.aps.org/doi/10.1103/PRXQuantum.2.040204.

[16] Philip Krantz et al. "A Quantum Engineer's Guide to Superconducting Qubits". In: *Applied Physics Reviews* 6.2 (2019), p. 021318. DOI: 10.1063/1.5089550.

[17] R.L. Kautz. "Jim Zimmerman and the SQUID". In: *IEEE Transactions on Applied Superconductivity* 11.1 (2001), pp. 1026–1031. DOI: 10.1109/77.919524.

[18] Alexandre Blais et al. "Cavity Quantum Electrodynamics for Superconducting Electrical Circuits: An Architecture for Quantum Computation". In: *Physical Review A* 69.6 (2004), p. 062320. DOI: 10.1103/PhysRevA.69.062320.

[19] Alexandre Blais et al. "Circuit Quantum Electrodynamics". In: *Reviews of Modern Physics* 93.2 (2021), p. 025005. DOI: 10.1103/RevModPhys.93.025005.

[20] José Chávez-Garcia et al. "Weakly Flux-Tunable Superconducting Qubit". In: (Mar. 2022). DOI: 10.48550/arXiv.2203.04164.

[21] Qiskit Metal Development Team. *Transmon Qubit CPB Hamiltonian Charge Basis*. Qiskit Metal Documentation. Accessed: 1 May 2026. URL: https://qiskit-community.github.io/qiskit-metal/tut/4-Analysis/4.34-Transmon-qubit-CPB-hamiltonian-charge-basis.html#Anharmonicity.

[22] J. A. Schreier et al. "Suppressing charge noise decoherence in superconducting charge qubits". In: *Phys. Rev. B* 77 (18 May 2008), 180502(R). DOI: 10.1103/PhysRevB.77.180502. URL: https://link.aps.org/doi/10.1103/PhysRevB.77.180502.

[23] QuEra Computing. *T1 Relaxation Time*. QuEra Glossary. Accessed: 7 May 2026. URL: https://www.quera.com/glossary/t1-relaxation-time.

[24] Mingrui Xu et al. "Radiative Cooling of a Superconducting Resonator". In: *Phys. Rev. Lett.* 124 (3 Jan. 2020), p. 033602. DOI: 10.1103/PhysRevLett.124.033602. URL: https://link.aps.org/doi/10.1103/PhysRevLett.124.033602.

[25] National Institute of Standards and Technology. *About Cryogenics*. NIST Cryogenic Technologies Group. Accessed: 28 April 2026. URL: https://trc.nist.gov/cryogenics/aboutCryogenics.html.

[26] National Institute of Standards and Technology. *Material Properties: OFHC Copper (UNS C10100/C10200)*. NIST Cryogenic Material Properties Database. Accessed: 14 May 2026. URL: https://trc.nist.gov/cryogenics/materials/OFHC%20Copper/OFHC_Copper_rev1.htm.

[27] National Institute of Standards and Technology. *Material Properties: 304 Stainless (UNS S30400)*. NIST Cryogenic Material Properties Database. Accessed: 14 May 2026. URL: https://trc.nist.gov/cryogenics/materials/304Stainless/304Stainless_rev.htm.

[28] Ray Radebaugh. "Pulse Tube Cryocoolers". In: (2003), pp. 1–15.

[29] "5 - Cooling Down to 0.3K". In: (2008). Ed. by Guglielmo Ventura and Lara Risegari, pp. 105–141. DOI: https://doi.org/10.1016/B978-008044479-6.50006-6. URL: https://www.sciencedirect.com/science/article/pii/B9780080444796500066.

[30] J. A. Bert. *Pulse Tube Refrigerators*. Stanford University, Physics 210. Accessed: 7 May 2026. 2007. URL: http://large.stanford.edu/courses/2007/ph210/bert2/.

[31] SHI Cryogenics Group. *RP-082B2 4K Pulse Tube Cryocooler Series*. Product specification page. Accessed: 3 March 2026. URL: https://shicryogenics.com/product/rp-082b2-4k-pulse-tube-cryocooler-series/.

[32] N. Almtireen, J. J. Brandner, and J. G. Korvink. "Pulse Tube Cryocooler: Phasor Analysis and One-Dimensional Numerical Simulation". In: *Journal of Low Temperature Physics* 199 (2020), pp. 1179–1197. DOI: 10.1007/s10909-020-02378-6.

[33] L. Piccirillo, G. Coppi, and A. May. *Miniature Sorption Coolers: Theory and Applications*. 1st. CRC Press, 2018. DOI: 10.1201/9781351188678. URL: https://doi.org/10.1201/9781351188678.

[34] Yusef Omidi Khaniabadi et al. "Adsorption of 4-chlorophenol from aqueous solution using activated carbon synthesized from aloe vera green wastes". In: *Journal of Advances in Environmental Health Research* 3 (Nov. 2015), pp. 1–10. DOI: 10.22102/jaehr.2015.40194.

[35] Sigurd Skogestad. *Introduction to Vapor/Liquid Equilibrium*. Chemical and Energy Engineering, Norwegian University of Science and Technology. Accessed: 14 May 2026. URL: https://skoge.folk.ntnu.no/septek/more-material/flash_from_skogestad_book.pdf.

[36] Frank Pobell. *Matter and Methods at Low Temperatures*. 3rd ed. Springer, 2007. DOI: 10.1007/978-3-540-46360-3.

[37] NumPy Developers. *numpy.trapezoid*. NumPy Documentation. Accessed: 14 May 2026. URL: https://numpy.org/doc/stable/reference/generated/numpy.trapezoid.html.

[38] Xi Xt et al. "Progress and Challenges of Sub-Kelvin Sorption Cooler and Its Prospects for Space Application". In: *Journal of Low Temperature Physics* 199 (June 2020). DOI: 10.1007/s10909-020-02442-1.

[39] Georgina Klemencic et al. "A continuous dry 300 mK cooler for THz sensing applications". In: *Review of Scientific Instruments* 87 (Apr. 2016), p. 045107. DOI: 10.1063/1.4945691.

[40] Wikipedia contributors. *Phase diagram of ^{3}He–^{4}He mixture*. Figure in: Dilution refrigerator. Accessed: 14 May 2026. URL: https://en.wikipedia.org/wiki/Dilution_refrigerator.